CSC126: Fundamentals of Algorithms and Computer Problem Solving

UNIVERSITI TEKNOLOGI MARA PAHANG

ASSESSMENT #1

QUIZ – 1 HOUR (10%)

Student Name: .....

Student ID: ..... Group: .....

Lecturer's Name: .....

CLO1 – Apply basic concepts and algorithms in computer programming.

SELF DECLARATION

1. I know that plagiarism is wrong. Plagiarism is to use other's work and pretend that it is one's own.
2. This quiz is my own work.
3. I have not involved, and will not allow anyone to copy my work with the intention of passing it off as their own work.
4. I acknowledge that copying someone else's work (or part of it) is wrong and declare that my quiz is my own work.

Answer ALL questions.

QUESTION 1

a. Differentiate between pseudocode and flowchart.

(4 marks)

b. Write a C++ statement to declare the following:

i. Declare a variable named dept_name as character. The length of variable is 50.
ii. Two float variables called x1 and x2 and initialize both to 0.0.

(4 marks)

c. Write a C++ statement that is equivalent to expression below:

$$C = \sqrt{\frac{a^2 + b^2 - 2ab}{4ac^2}}$$

(2 marks)

QUESTION 2

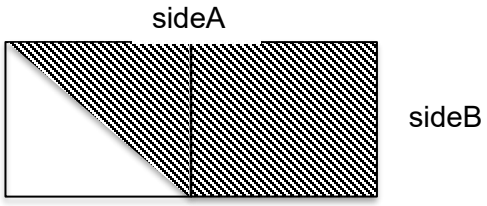
i. Construct a flowchart from the pseudocode below to read and display the following variables.

```
Start
    Read mark1, mark2, mark3
    Display total_marks,average
End
```

(5 marks)

ii. Based on the following diagram:

You are assigned to write a program to determine the area for the following diagram. Prompt the user to input two integers **sideA** and **sideB**, then calculate and display the area of the shaded area.



i. Write the analysis of this program.

(5 marks)

ii. Draw the flowchart for the above problem.

(5 marks)

iii. Write a C++ program segment for the given problem.

(5 marks)